Experiment-20

Aim

To perform image sharpening using a Laplacian mask with a negative center coefficient and observe the enhancement of edges and fine details in the image.

Procedure

1. Select and load the input image.
2. Convert the image into grayscale.
3. Define the Laplacian mask:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

4. Apply the mask to the image using convolution.
5. The Laplacian operation detects rapid intensity changes and extracts edges.
6. Since the mask has a **negative center coefficient**, add the Laplacian result to the original image to obtain the sharpened image:

$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$

7. Display the original image and the sharpened image.
8. Observe that edges and fine details become more prominent.

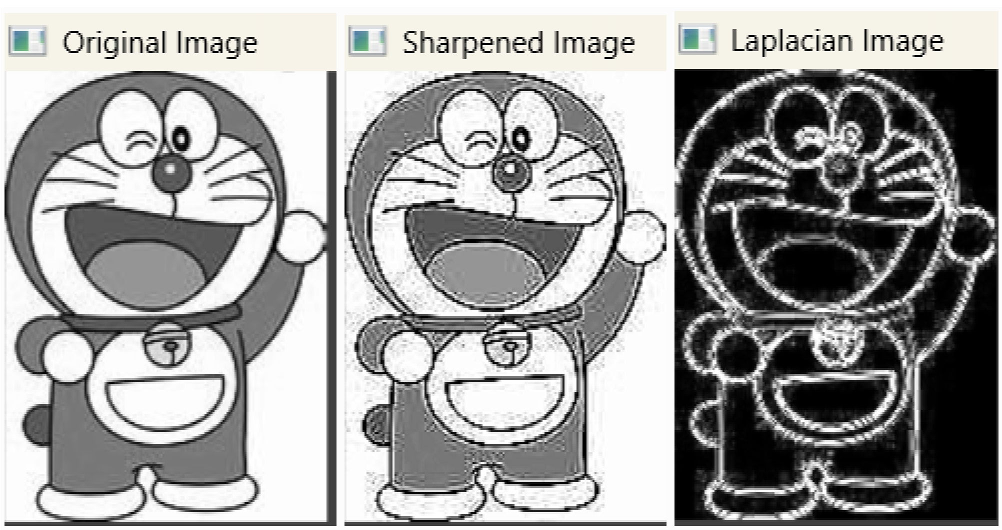
Python Code

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
mask = np.array([
    [0, 1, 0],
    [1, -4, 1],
```

```
    [0, 1, 0]
], dtype=np.float32)
laplacian = cv2.filter2D(gray, cv2.CV_32F, mask)
sharpened = gray.astype(np.float32) - laplacian
sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)
cv2.imshow("Original Image", gray)
cv2.imshow("Laplacian Image", cv2.convertScaleAbs(laplacian))
cv2.imshow("Sharpened Image", sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:

Output:



Results:

The image was successfully sharpened using the Laplacian mask with a negative center coefficient. The sharpened image shows enhanced edges, boundaries, and fine details compared with the original image.